Category

has
- objects
- morphisms

statisfies
- composition
- asociativity
- identity

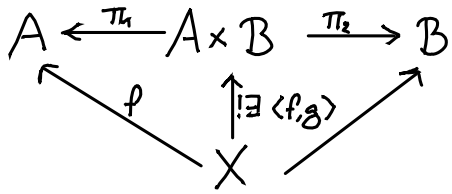
Terminal object 1

$\forall A \in Ob\mathcal{C}. \exists! \; !_A : A \to 1.$

universal property for product

The product of two objectct A and B is

1. An object $A \times B$
2. Two projection morphisms $\pi_1 : A \times B \to A$, $\pi_2 : A \times B \to B$, such that $\forall X \in Ob\mathcal{C}. \forall f : X \to A. \forall g : X \to B.$
$\exists! \langle f, g \rangle : X \to A \times B. \; \pi_1 \circ \langle f, g \rangle = f \wedge \pi_2 \circ \langle f, g \rangle = g$



Category with finate products has

1. Terminal object
2. Binary product $A \times B$ for every pair A, B

Properties

- uniguness up to isomorphism
(There exist meny products, but they are isomorphic)
- $A \times B \cong B \times A$
- $(A \times B) \times C = A \times (B \times C)$
- $A \times 1 = A$

A group object

has
- an object G
- morphisms
  1. $m : G \times G \longrightarrow G$
  2. $i : G \longrightarrow G$
  3. $e : 1 \longrightarrow G$

satisfying

- associativity $m \circ (m \times id_G) = m \circ (id_G \times m)$

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{m \times id_G} & G \times G \\ \downarrow{\scriptstyle id_G \times m} & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow{m} & G \end{array}$$

- identity law $m \circ (e \times id_G) = \lambda_G \quad m \circ (id_G \times e) = \rho_G$

where $\lambda_G : 1 \times G \longrightarrow G$ and $\rho_G : G \times 1 \longrightarrow G$

$$\begin{array}{ccccc} 1 \times G & \xrightarrow{e \times id_G} & G \times G & \xleftarrow{id_G \times e} & G \times 1 \\ & {\scriptstyle \lambda_G} \searrow & \downarrow{\scriptstyle m} & \swarrow {\scriptstyle \rho_G} & \\ & & G & & \end{array}$$

- inverse law $m \circ (i, id_G) = e \circ !_G = (i, id_G) \circ m$

where $\Delta_G = \langle id_G, id_G \rangle : G \longrightarrow G \times G$

$$\begin{array}{ccccc} & & G & & \\ & {\scriptstyle (i, id_G)} \swarrow & \downarrow{\scriptstyle !_G} & \searrow {\scriptstyle (id_G, i)} & \\ G \times G & & 1 & & G \times G \\ & {\scriptstyle m} \searrow & \downarrow{\scriptstyle e} & \swarrow & \\ & & G & & \end{array}$$

Category $\mathbf{Grp}_{\mathcal{C}}$ has

- objects: Group objects
- morphisms: $(G, m, e, i) \longrightarrow (G', m', e', i')$

such that

$$\begin{array}{ccc} G \times G & \xrightarrow{h \times h} & G' \times G' \\ \downarrow m & & \downarrow m \\ G & \xrightarrow{h} & G' \end{array}$$

$$\begin{array}{ccc} & 1 & \\ {\scriptstyle e}\swarrow & & \searrow{\scriptstyle e'} \\ G & \xrightarrow{h} & G' \end{array}$$

$$\begin{array}{ccc} G & \xrightarrow{h} & G' \\ \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i'} \\ G & \xrightarrow{h'} & G' \end{array}$$

comutes

Algebraic signature

is a tuple (S, Ω, ar)

- S... Sorts: A set of types (or sorts)
- Ω.. Operation symbols: A set of operation names
- ar... Arity function: A mapping that assigns $f \in \Omega$ its input and output type

(if $|S| = 1$ then arity represents number of inputs)

Terms X is a set of variables

- variable $x \in X$ is a term
- if $f \in \Sigma$ is an n-ary operation symbol and $t_1 \ldots t_n$ are terms, then $f(t_1 \ldots t_n)$ are terms

Equation over a signature Σ is a pair (t_1, t_2). They state, that two different term trees must be treated as equal when evaluated in a model.

Algebraic theory consists of

- signature Σ
- a set of equations

$T = (\Sigma, \mathcal{E})$

T-Object in category with finite products is

- object A
- for each n-ary operation $f \in \Sigma$ a morphism $A^n \longrightarrow A$, such that all equations in $\mathcal{E}$ are true in A

Lawvere theory is a category $\mathbb{T}$ with

- objects: $\underline{n}$ $\forall n \in \mathbb{N}$
- morphisms: $\underline{m} \longrightarrow \underline{n}$: n-tuples
  $(t_1(x_1 \dots x_n), \dots, t_m(x_1 \dots x_n))$
  - composition: substitution
  - identity $n \to n$: tuple $(x_1 \dots x_n)$
- terminal object: $\underline{m} \times \underline{n} = \underline{m+n}$

Mod$(T, \mathbb{C})$ ... category of T-objects in $\mathbb{C}$

- objects: T-objects
- morphisms: $f : A \to B$ such that square

$$\begin{array}{ccc} A^n & \xrightarrow{f^n} & B^n \\ \downarrow{\scriptstyle \sigma_A} & & \downarrow{\scriptstyle \sigma_B} \\ A & \xrightarrow{f} & B \end{array}$$

commutes for all σ of arity k.

FP$[\mathbb{T}, \mathbb{C}]$ ... category of finite product preserving func.

- objects: functor $F : \mathbb{T} \longrightarrow \mathbb{C}$ such that
  - $F(\underline{0})$ = terminal object in $\mathbb{C}$
  - $F(\underline{m+n}) = F(\underline{m}) \times F(\underline{n})$

  (The functor is determined by $F(1) = A$)
- morphisms: natural transformation

  $\alpha : F \Rightarrow G$, so that square

a family of morphisms ↑

$$\begin{array}{ccc} F(n) & \xrightarrow{\alpha_n} & G(n) \\ \downarrow{\scriptstyle F(t)} & & \downarrow{\scriptstyle G(t)} \\ F(m) & \xrightarrow{\alpha_m} & G(m) \end{array}$$

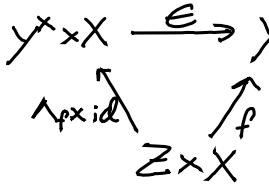
Theorem: **Mod$(T, \mathbb{C}) \simeq$ FP$[\mathbb{T}, \mathbb{C}]$**

Cartesian closed category

- has terminal object
- any two objects X, Y have a product $X \times Y$ in $\mathbb{C}$
- any two objects X, Y have exponential Y^X

Exponential object

Y^X is object with a morphism: $\epsilon : Y^X \times X \longrightarrow Y$ such that for any morphism $f : Z \times X \longrightarrow Y$ there exists a unique morphism $\Lambda f : Z \longrightarrow Y^X$ such that diagram commutes



- exponentials are unique up to isomorphism.
- X is exponentiable if Y^X exists $\forall Y \in |\mathbb{C}|$
- X is exponentiable $\Rightarrow$ $(-)^X$ is a functor
- X is exponentiable $\Leftrightarrow$ the functor $(-) \times X$ has a right adjoint.

Natural transformation η from F to G, where $F, G : \mathbb{C} \longrightarrow \mathbb{D}$ are functors, is a family of morphisms $\eta_X : F(X) \longrightarrow G(X)$, such that

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ \downarrow{\scriptstyle F(f)} & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow{\eta_Y} & G(Y) \end{array}$$

commutes.

Right adjoint of a functor $F : \mathbb{C} \longrightarrow \mathbb{D}$ is a functor $G : \mathbb{D} \longrightarrow \mathbb{C}$ such that

$\forall C \in \mathbb{C}.\ Hom_{\mathbb{D}}(F(C), D) \cong Hom_{\mathbb{C}}(C, G(D))$.

So that $Hom_{\mathbb{D}}(F(_), D) \cong Hom_{\mathbb{C}}(_, GD))$ is a natural isomorphism

λ-theory $T(\text{Base}, \Sigma, \mathcal{E})$

- Base ... set of base types
- Σ ... signature with elements $c:A$ such that each element has exactly one type
- $\mathcal{E}$ ... a set of typed equations $\Gamma \vdash s = t : A$ between terms $\Gamma \vdash s : A$ and $\Gamma \vdash t : A$

Simply typed λ-calculus

- Base ... a set of base types
- types ... $A, B ::= a \in \text{Base} \mid A \times B \mid 1 \mid A \to B$
- terms ... $\Gamma \vdash t : A$
- context Γ ... $x_1 : A_1, \ldots, x_n : A$, such that $i \neq j \Rightarrow x_i \neq x_j$

Pravila:

- $$\frac{}{x_1:A_1 \ldots x_n:A_n \vdash x_i : A_i}$$

equality: $\dfrac{\Gamma \vdash t : A}{\Gamma \vdash t = t : A}$ + symmetry + transitivity

congruence: $\dfrac{\Gamma \vdash s = s' : A \quad \Gamma \vdash t = t' : B}{\Gamma \vdash (s,t) = (s',t') : A \times B}$ $\qquad$ $\dfrac{\Gamma, x:A \vdash t = t' : B}{\Gamma \vdash \lambda x.t = \lambda x.t' : A \to B}$

- β-rules
  - $\dfrac{\Gamma \vdash t_1 : A_1 \quad \Gamma \vdash t_2 : A_2}{\Gamma \vdash \pi_i(t_1, t_2) = t_i : A_i}$
  - $\dfrac{\Gamma, x:A \vdash s : B \quad \Gamma \vdash t : A}{\Gamma \vdash (\lambda x:A.\, s)\, t = s[t/x] : B}$
- η-rules

$$\frac{\Gamma \vdash u : 1}{\Gamma \vdash u = * : 1} \qquad \frac{\Gamma \vdash u : A \times B}{\Gamma \vdash u = (\pi_1 u, \pi_2 u) : A \times B}$$

$$\frac{\Gamma \vdash u : A \to B}{\Gamma \vdash u = \lambda x{:}A.\, u\, x : A \to B}$$

T-model in CCC

- objects: $a^{\mathbb{C}}$ for all $a \in Base$
  - $(A \times B)^{\mathbb{C}} = A^{\mathbb{C}} \times B^{\mathbb{C}}$
  - $1^{\mathbb{C}} = \underline{1}$
  - $(A \to B)^{\mathbb{C}} = A^{\mathbb{C}} \longrightarrow B^{\mathbb{C}}$
- arrows: $\Gamma^{\mathbb{C}} \xrightarrow{\Gamma \vdash t : A} A^{\mathbb{C}}$

  where $()^{\mathbb{C}} = \underline{1}$

  $(\Gamma, x:A)^{\mathbb{C}} = \Gamma^{\mathbb{C}} \times A^{\mathbb{C}}$
- sellbject to condition

  $\Gamma \vdash s = t : A \implies (\Gamma \vdash s : A)^{\mathbb{C}} = (\Gamma \vdash t : A)^{\mathbb{C}}$

  constants: $1 \longrightarrow A^{\mathbb{C}} \quad \forall c : A \in \Sigma$

Theorem:

$\Gamma \vdash_T s = t : A \Leftrightarrow \forall \mathbb{C} \in CCC.\ \forall$ T-model in $\mathbb{C}.\ (\Gamma \vdash s = t : A)^{\mathbb{C}}$

Every derivable equation in the model is provable and every provable equation is derivable.